# Infosys - Coding 17

HackerRank

You are given a grid of integers with N rows and M columns. Your task is to select exactly one cell from every row. For each row i (from 0 to N−1), let the column index of the cell you select be denoted as c[i].

Your selection must adhere to a specific constraint regarding the range of columns used: the difference between the maximum column index selected across all rows and the minimum column index selected across all rows must be less than or equal to a given integer K. That is, max(c[i]) − min(c[i]) ≤ K.

The total score is calculated as:

1. Sum_of_selected_cells = Σ grid[i][c[i]] for i from 0 to N−1.
2. Spread = max(c[i]) − min(c[i]).
3. Penalty_Value = P × Spread.
4. Total_Score = Sum_of_selected_cells − Penalty_Value.

Find the maximum possible total score achievable while satisfying the column range constraint.

**Input Format**

Line 1: integer N (rows).

Line 2: integer M (columns).

Line 3: integer K (max allowed column index difference).

Line 4: integer P (penalty constant).

Next N lines: M space-separated integers, row i of grid.

**Constraints**

1 <= N <= 100

1 <= M <= 100

0 <= K <= 100

0 <= P <= 1000

-10^9 <= grid[i][j] <= 10^9

**Output Format**

The output contains an integer value

**Sample Input 0**

```
2
3
```

```
1
10
10 1 5
2 20 3
```

**Sample Output 0**

```
21
```

**Explanation 0**

Selecting column 1 from both rows gives 1 (row 0) + 20 (row 1) = 21 with spread 0, so no penalty. Total score = 21.

**Sample Input 1**

```
3
4
0
5
10 2 8 1
5 15 3 7
12 4 6 9
```

**Sample Output 1**

```
27
```

**Explanation 1**

Since K = 0, all selected cells must be in the same column. Column 0 yields the highest sum 10 + 5 + 12 = 27, with spread 0 and no penalty. Total score = 27.